(UV/Vis & IR spectroscopy)

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Organic Chemistry (CHEM-317)

Q. Write the application of IR spectroscopy?

Ans Infrared spectroscopy is a versatile analytical technique with numerous applications in various fields.

1: chemical analysis:-

IR spectroscopy is widely used to identify the functional groups present in organic compounds, aiding in structural determination

It can be used for quantitative analysis of the concentration of specific compounds in a mixture.

2: Pharmaceutical and health care:-

IR spectroscopy helps in studying the composition of pharmaceutical formulations, ensuring quality and consistency.

It is used for the analysis of proteins, nucleic acids and other biomolecules.

3: material science:-

IR spectroscopy is essential for characterizing polymers.

It is used in geology and

mineralogy for the identification of minerals and the study of mineral propertites.

4: Environmental analysis:-

IR spectroscopy can be used to monitor air pollutants.

It helps in assessing the quality of water by detecting contaminants and pollutants.

5: Forensics:-

IR spectroscopy aids in forensic science for the analysis of trace evidence.

It can be used in the examination of inks, papers and documents for authenticity.

6: Food industry:-

IR spectroscopy can be employed to assess the quality of food products and to detect contaminants.

It helps in determing the nutritional content of food items

7: Chemical manufacturing and Process control:-

IR spectroscopy is used

to monitor chemical processes and ensure product quality. It can track chemical reactions in real time.

8: Art and cultural Heritage conservation:-

IR spectroscopy is used to analyze pigments, binders, and other materials in artworks to aid in restoration and preservation.

9: Biotechnology:-

IR spectroscopy is used to study protein secondry structure and conformation, aiding in drug development and protein research.

10: Remote sensing:-

IR spectroscopy is used in satellite-based remote sensing to study atomsphere composition and environmental changes.

In summary, It is a valuable tool for chemical analysis and material characterization.

Q Write the instrumentation and sample handling of IR spectroscopy?

Ans IR spectroscopy is a technique to study the rotational and vibrational modes of molecules.

~~* Instrumentation:~~

* Sample handling:

1: Gases:-

The gaseous sample is introduced into a "gas cell". The wall of cells are made of sodium chloride. Sodium chloride windows allow the cell to be placed directly in the path of sample beam. The gas cell is usually 10cm long. very few organic compounds can be examined as gases.

2: Solution:-

It is most conveniet to determine the spectrum in solution. Excellent solvents are those which have poor adsorption of its own. Unlike UV-Vis spectroscopy, too many

Solvents cannot be used in this technique. In IR technique, all solvents do absorb in one region or the other. Some important solvents which may be used are:

(i) Chloroform (ii) Carbon tetrachloride

(iii) Carbon disulphate

Water cannot be used as solvent as it absorbs in several regions of the IR spectrum.

3: Liquid:

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4: Solids:

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★ Instrumentation:-

A simple IR spectrophotometer typically consists of the following basic instrumentation components.

1: Infrared light source:

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2: (4) Monochromator:

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3: (2) Sample and Refrence beams:

Light from the source is split

into two beams. One of the beams is passed through the sample under examination. It is called the sample beam. The other beam is called the refrence beam. When the beam passes through the sample, it become less intense due to the absorption of certain frequency. These beams directed to the chopper.

4: Chopper:-
(3)

The two beams are made to fall on a segmented mirror M called chopper. This is done with the help of two mirror M_1 and M_2. The chopper rotates at a definate speed and reflects the both beams to a monochromator grating. →

5: Detector:

→ [As grating rotates slowly, it sends individual to the detector]

→ The detector measure the intensity of the transmitted or absorbed infrared light, after it has interacted with beam

Commonly detector for IR spectroscopy include thermocouple, bolometer or MCT detector.

6: Amplifier:

Amplifier increase the sensitivity of the detector and improving the quality of the acquired data by enhancing the signal, reducing noise and providing a wider dynamic range.

7: Recording and Analysis:

Amplified data is then recorded and processed by the instrument's software, leading to the generation of an accurate IR spectrum that can further analyzed and interpreted by chemists

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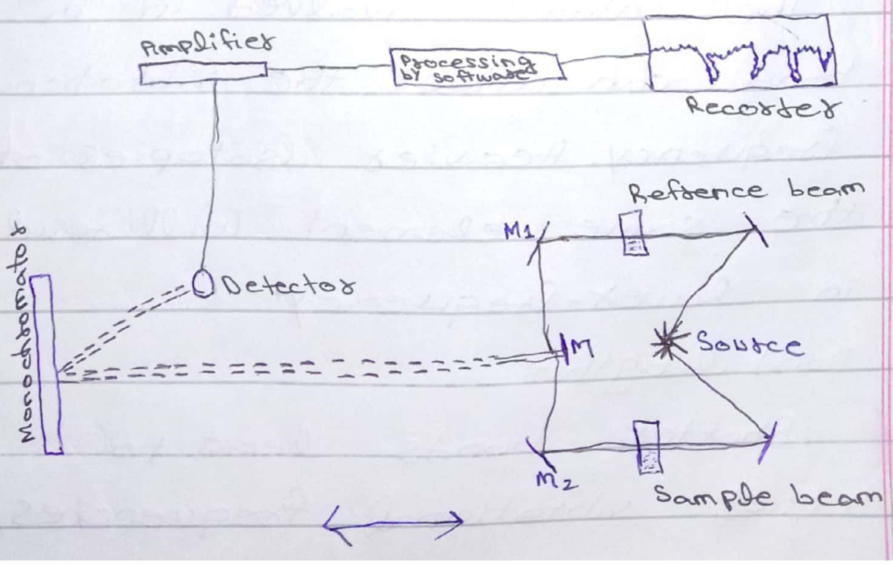

Q Discuss factors influencing the vibrations in IR spectroscopy?

Ans Several factor that influence the vibrations in IR spectroscopy:

1: Molecular Mass:-

★ Heavier molecules tend to vibrate at lower frequencies, while lighter molecules vibrate at lower frequencies. This is due to relationship between mass and vibrational energy.

2: bond strength:-

★ Stronger chemical bonds results in higher vibrational frequencies. So:

$$-C\equiv C- \; > \; >C=C< \; > \; -\overset{|}{\underset{|}{C}}-\overset{|}{\underset{|}{C}}-$$

3: Atomic mass:-

★ The atoms involved in a bond can affect the vibrational frequency. Heavier isotopies of the same element will result in lower-frequency.

4: Bond length:-

★ Shorter bonds lead to higher vibrational frequencies, as they are stiffer and have higher force constants.

5: Dipole moment:-

* molecules with a greater dipole moment experiance stronger interactions with the electric field of the IR radiation, leading to more intense IR absorptions.

6: Electronegativity:-

Differences in electronegativity between bonded atoms affect the strength of the bond and consequently, the vibrational frequency.

7: Conjugation:-

In conjugated systems, such as in aromatic compounds, the distribution of electron density can affect the vibrational frequencies of bonds.

8: Steric Effects:-

Bulky subsitusents can hinder or alter the vibrational modes in a molecule, leading to shifts in the IR spectrum.

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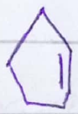
9: Hydrogen Bonding:-
Hydrogen bonds affect vibrational frequencies, often resulting in characteristic shifs in the spectrum.

10 Solvent Effects:-
The nature of the solvent can influence the vibrational frequencies observed in the IR spectrum, especially for polar molecules.

Q Write down the applications of UV/Vis spectroscopy?

Ans From Notes & GRC Page no. 536

Q How would each of following pairs group differ in their IR spectra?

Ans
(a)  and $HC \equiv CCH_2CH_2CH_3$

$C=C$ bond
$1650\ cm^{-1}$
Csp^2-H
$3150-3000\ cm^{-1}$

$C \equiv C$ bond
$2250\ cm^{-1}$
$Csp-H$
$3300\ cm^{-1}$

(d) $HC\equiv C-CH_2-N(CH_2CH_3)_2$ Csp-bond $3300\,cm^{-1}$

and

$CH_3(CH_2)_5C\equiv N$

No Csp–H bond

(e) $CH_3-C\equiv C-CH_3$ and $CH_3-CH_2-C\equiv CH$

No absorption due to symmetry

Csp – H bond

$3300\,cm^{-1}$

$C\equiv C$ bond

$2250\,cm^{-1}$

(c) $CH_3CH_2-\underset{\underset{O}{\|}}{C}-CH_3$ and $CH_3-CH=CH-CH_2OH$

C=O bond

$1700\,cm^{-1}$

O–H bond

$3200-3600\,cm^{-1}$

(b) $CH_3CH_2-\overset{\overset{O}{\|}}{C}-OH$ and $CH_3-\overset{\overset{O}{\|}}{C}-OCH_3$

O–H bond

$> 3000\,cm^{-1}$

No O–H bond

$\longleftrightarrow$